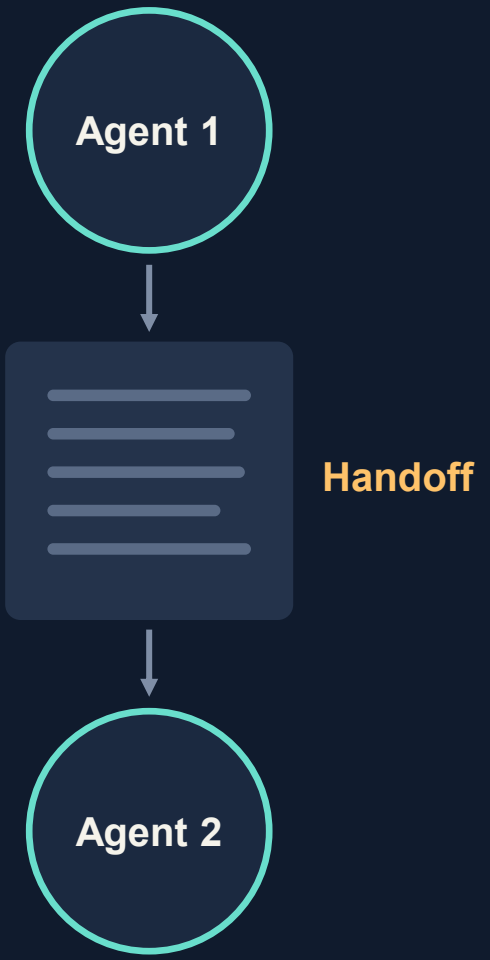
CS3227 AGENTIC SOFTWARE ENGINEERING · WEEKS 8 AND 9

# Free-form summaries vs structured checklists

Which handoff format loses less information between AI agents?

*An experiment on a real MP3 task*



WHY THIS EXPERIMENT

# Our MP3 agents pass work along in handoffs

In a multi-agent workflow (Week 8), each agent ends its task by handing off to the next agent. The next agent starts fresh, so the handoff is all it knows.

OUR MP3 WORKFLOW, FOR EACH FEATURE




Each handoff uses a fixed checklist. If it drops something important, all subsequent agents work without it.

GOAL

**Find practical ways to retain context when agents hand off work.**

This experiment tests one choice: how the handoff is written.

WHY IT IS HARD

# A handoff squeezes a long history into a short note

Week 9 calls this context compaction, and warns that compaction is lossy: whatever the handoff leaves out, the next agent never sees.




**Message history**
every message, tool call and tool result
3,449 to 3,551 words in this experiment

**Handoff**
200-word limit

**Starts fresh**
with only the handoff

THE QUESTION

# Which handoff format keeps more information?

I compared 2 ways of writing the same handoff. Both were told to cover the same 7 topics, from the same message history, in at most 200 words.

### FREE-FORM SUMMARY

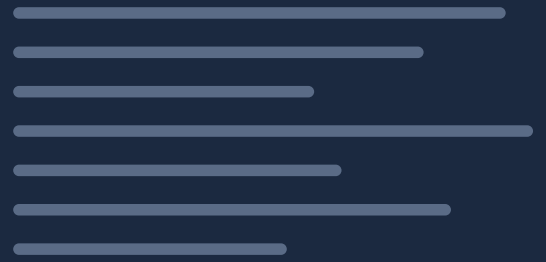
The 7 topics as connected prose, with no headings or bullets



### STRUCTURED CHECKLIST

The 7 topics as fixed fields, like our MP3 handoffs

- **Agent and Status**
- **Inputs**
- **Changed**
- **Checks**
- **Decisions needed**
- **Assumptions**
- **Untrusted content seen**



SAME WORD LIMIT, BUT THE FORMAT CAN CHANGE WHAT THE WRITER INCLUDES

**A summary could help:** no words go on labels, leaving room to explain why.

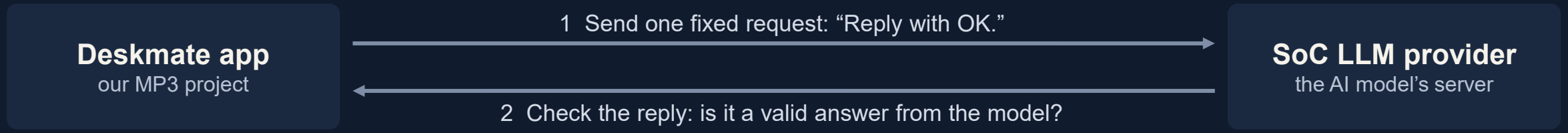
**A checklist could help:** each field reminds the writer to cover that topic.

Then I measured **how much information the next agent correctly recovers, misses or gets wrong.**

THE CODING TASK

# The task: fix 5 bugs in Deskmate’s LLM health check

First, a coding agent did a real MP3 task. Deskmate’s health check tests whether our app can reach the SoC LLM provider, and its starting code had 5 bugs.




## 5 BUGS IN THE STARTING CODE

1

### Timeout

No overall 10-second limit on the whole check

2

### Proxy

May send the request via the machine’s proxy server

3

### Extra header

Sends a User-Agent header it should leave out

4

### Bad access tokens

Some crash the check instead of being refused

5

### Partial replies

A network reply cut off mid-transfer crashes the check

The fix also had to follow all other documented rules, such as which replies count as a success.

THE SETUP

# 3 agents pass the work along

The coding agent's message history goes to 2 handoff agents, one per format. A fresh receiving agent then answers 6 questions using only one handoff.




**5 coding runs × 4 handoff models × 2 formats = 40 handoffs**, each quizzed once

The 4 handoff models: GPT-5.6 Sol, GPT-5.6 Luna, Claude Opus 5.5 and Claude Sonnet 5.5

SCORING

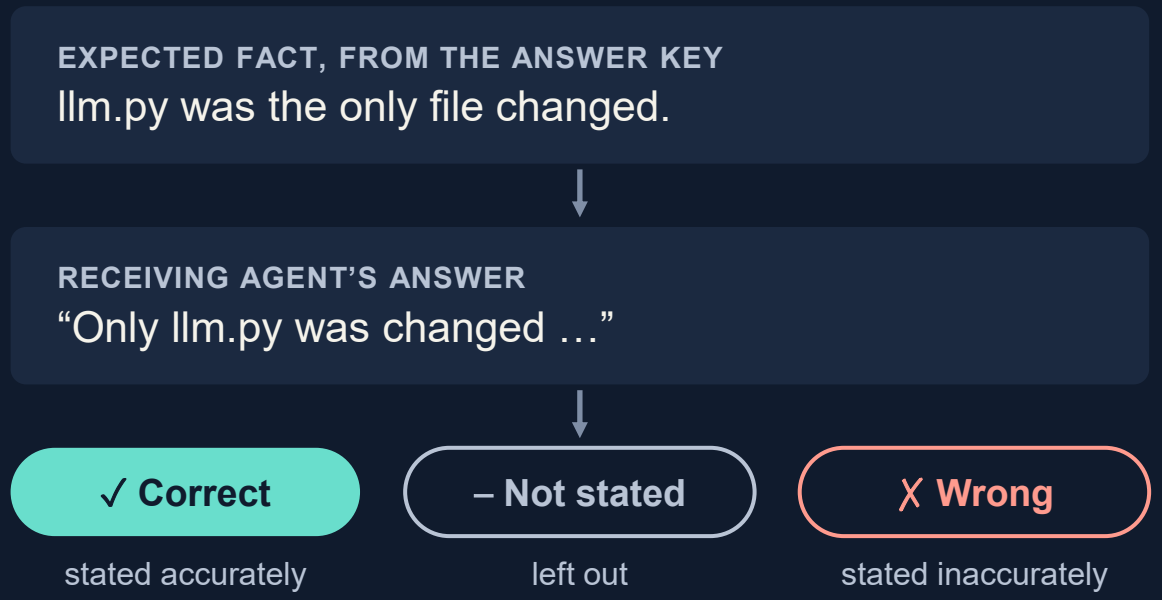
# Scoring: how many expected facts survived?

For each coding run, I wrote an answer key before its handoffs existed. It splits the 6 questions into expected facts, taken only from that specific run's message history.

## THE 6 QUESTIONS

1. Which files were changed?
2. Which checks were run, and what happened?
3. What did the agent leave unverified?
4. How and why was the timeout fixed?
5. Which provider responses count as a success?
6. What was recorded about proxies and headers?

## HOW ONE FACT IS SCORED



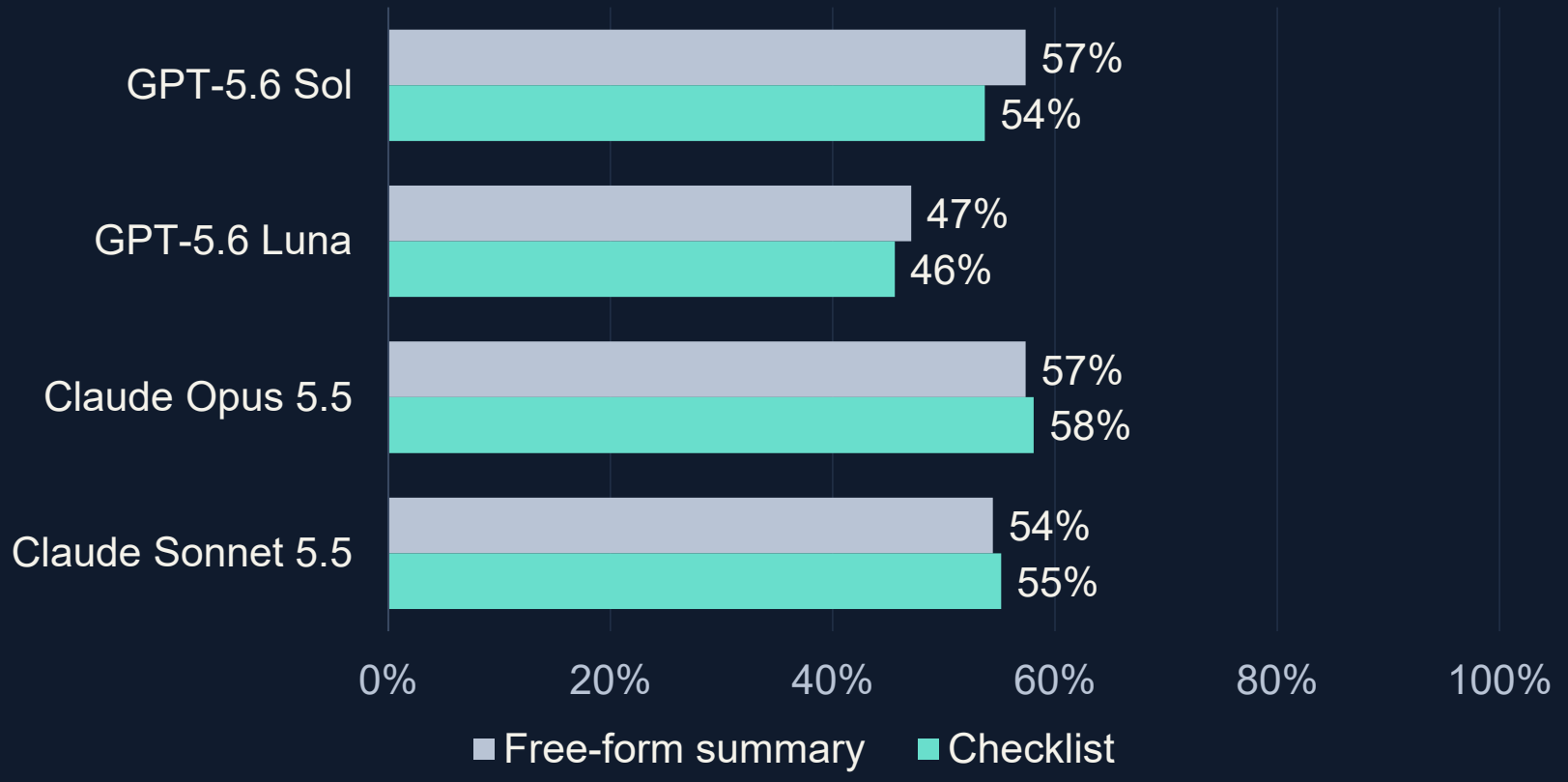

**Score = correct facts ÷ expected facts**

27 or 28 facts per run: 136 across all 5 runs, for each model and format

RESULT 1

# GPT leaned free-form, Claude leaned checklist

No format won overall. GPT models did slightly better with free-form summaries, and Claude models with checklists. Every gap was 5 facts or fewer, out of 136 across all 5 runs.




**GPT: summary ahead**
by 5 facts (Sol) and 2 facts (Luna)

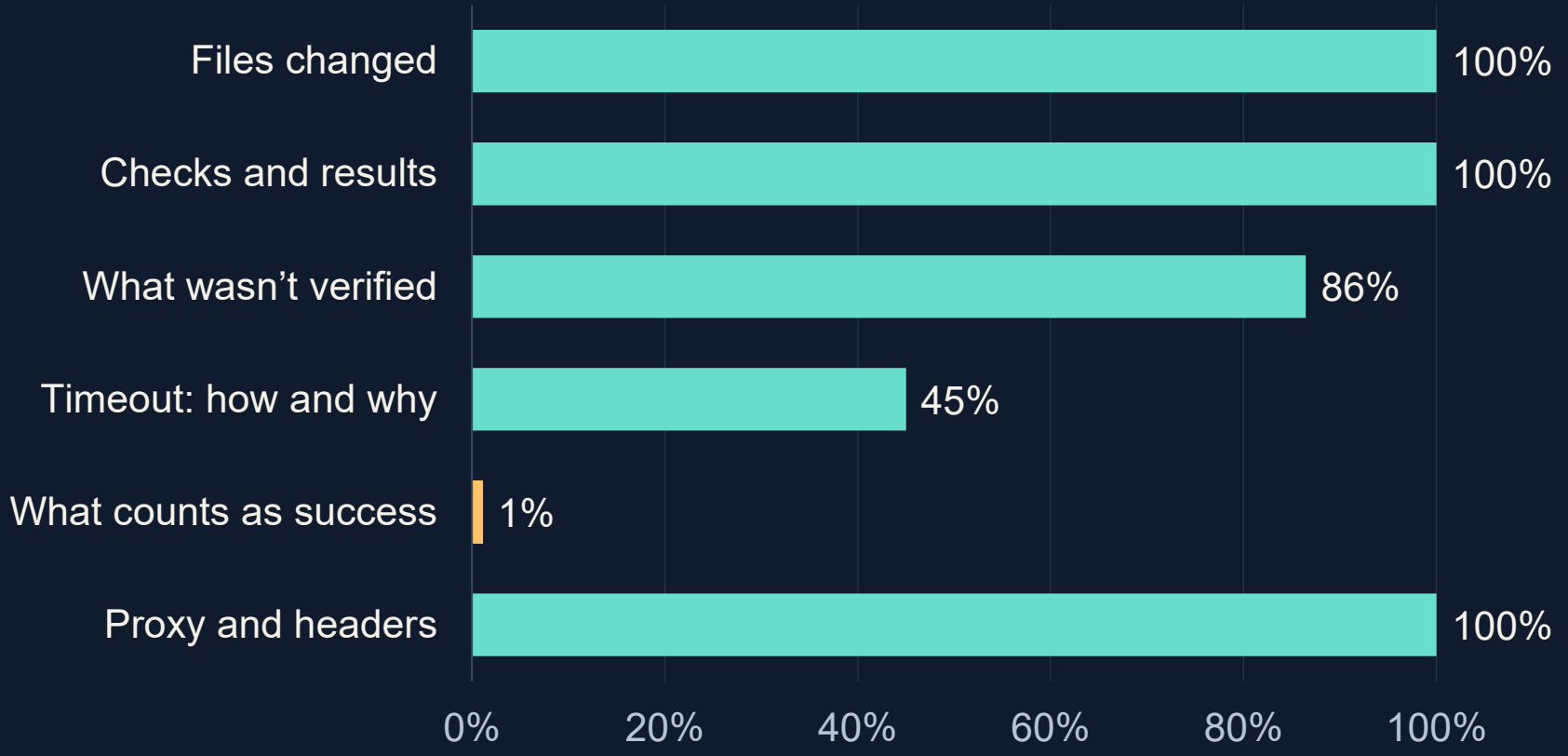
**Claude: checklist ahead**
by just 1 fact each (Opus and Sonnet)

**Stronger model, higher score**
In both formats, Sol beat Luna by 11 to 14 facts, and Opus beat Sonnet by 4 facts.

RESULT 2

# The real gap was in which facts survived

Facts about what the agent did nearly always survived. The rules for what counts as a successful response almost never did.




## 4 of 360

facts about what counts as success survived, across all 40 handoffs

**Likely reasons**

None of the 7 topics asked for these rules.

The rules weren’t among the 5 bugs, and handoffs mostly described changes.

Not the word limit: handoffs over 200 words left them out too.

AN EXAMPLE

# The checklist kept the test results but lost the rules

This real checklist reported that all 30 tests passed. But it never said which responses count as a success, so the receiving agent answered “not stated”.

### THE HANDOFF

GPT-5.6 Sol checklist, coding run 4

**Checks:** Offline standalone suite passed all 30 tests in 0.030s. Covered request shape/headers, tokens, response validation, 65,536-byte limit, …

✓ **Test results kept**

✗ **Nothing on which responses count as a success**

### THE RECEIVING AGENT

**Asked**

“Which provider responses count as a successful probe?”

**Answered**

*“not stated. The handoff mentions only response validation and shape validation, without saying which responses count as success.”*

### THE ANSWER KEY

9 expected facts for this question

- ✗ Status code is 200
- ✗ Body is valid UTF-8
- ✗ Body is valid JSON
- ✗ Has a non-empty “choices” list
- ✗ First choice has a text message
- ✗ Message is not blank
- ✗ Message need not say “OK”
- ✗ Content-Type header is ignored
- ✓ **Body is at most 65,536 bytes**

**1 of 9 recovered**

LIMITATIONS

# What this experiment can’t tell us

These results describe this one setup. They don’t show that either format is better in general.

### </> 1 coding task

And a quiz, not real follow-up work.

### 1× No replicates

Each model wrote each handoff once, so randomness could explain small gaps.

### ≈ The formats ended up alike

Most free-form summaries still used labels like “Inputs:”, just within a paragraph.

### 200+ Some went over 200 words

12 of 40 handoffs, including 9 of the 10 Claude checklists.

### 1 1 reader model

Claude Opus 5.5 read every handoff, which may suit Claude-written ones.

### AI Graded by AI

An AI marked every fact, so some marks may be wrong.

TAKEAWAY

# Decide what goes into a handoff, not just how it looks

The format made only small differences. What mattered more was whether the handoff had the right facts.

## Lost information can’t be recovered. Extra information can be trimmed later.

Our handoffs kept what the agent did, but lost the rules the work had to meet.

IDEAS TO TEST NEXT, FOR OUR MP3 WORKFLOW

1 **Add a goal and success-criteria field**

**Structured note taking (Week 9):** the lecture’s note template starts with the Goal; our checklist has no such field.

2 **Point to the requirements**

**Just-in-time retrieval (Week 9):** link the contract, so the next agent can look the rules up itself.

*Neither idea was tested in this experiment.*

BACKUP

# Full results by model and format

Correct facts out of 136 expected facts, across all 5 coding runs. No expected fact was scored wrong; the rest were not stated.

| HANDOFF MODEL | FORMAT | CORRECT FACTS | SHARE | OVER 200 WORDS |
|---|---|---|---|---|
| **GPT-5.6 Sol** | Free-form summary | **78 of 136** | 57% | 0 of 5 |
| **GPT-5.6 Sol** | Checklist | **73 of 136** | 54% | 0 of 5 |
| **GPT-5.6 Luna** | Free-form summary | **64 of 136** | 47% | 0 of 5 |
| **GPT-5.6 Luna** | Checklist | **62 of 136** | 46% | 0 of 5 |
| **Claude Opus 5.5** | Free-form summary | **78 of 136** | 57% | 3 of 5 |
| **Claude Opus 5.5** | Checklist | **79 of 136** | 58% | 5 of 5 |
| **Claude Sonnet 5.5** | Free-form summary | **74 of 136** | 54% | 0 of 5 |
| **Claude Sonnet 5.5** | Checklist | **75 of 136** | 55% | 4 of 5 |

BACKUP

# How the test was kept fair

### Same start for every coding run

A fresh copy of the same starting code, contract and 30 offline tests. All 5 repairs passed all 30 tests.

### Separate agents, no peeking

Every agent started in a fresh session. No agent saw the other handoff or the answer key.

### Fixed before the handoffs

Questions, prompts and each run’s answer key were frozen before that run’s handoffs were written.

### Isolated and audited

The coding agent could only read its 3 files, rewrite llm.py and run the tests, with no network or other commands. Records of all 110 scored model calls showed no other tool use.